

Working with Apache SINGA, the Deep Learning Library

Here's a short tutorial on how to install and configure Apache SINGA, the deep learning library that has been built for training machine learning models.



Apache SINGA is an open source deep learning library developed for training large-scale machine learning models efficiently across distributed systems. It is part of the Apache Software Foundation and focuses on scalability, flexibility, and ease of use for both research and production environments. It involves setting up dependencies, building from source (or installing via package managers), and configuring it for your specific use case (e.g., single-node or distributed training).

Why Apache SINGA?

Distributed training capabilities: SINGA excels in distributed training across multiple GPUs/nodes, making it ideal for

large-scale datasets or models (e.g., deep neural networks, transformers). It supports both data parallelism and model parallelism, and integrates with communication backends like MPI, gRPC, and NCCL for efficient inter-node coordination.

Fault tolerance: It automatically recovers from node failures during distributed training, ensuring robustness in production environments.

Its key features are listed below.

Horizontal scaling: Efficiently distributes training across multiple nodes using both data parallelism (splitting data) and model parallelism (splitting model layers).

Synchronous/asynchronous training: Supports flexible synchronisation strategies for distributed environments.

Framework	Scalability	Ease of Use	ONNX Support	Distributed Focus
SINGA	★★★★★	★★★	✓	Core Feature
PyTorch	★★★	★★★★★	✓	Requires add-ons
TensorFlow	★★★★★	★★★★★	✓	Requires TFX

Figure 1: Comparison with PyTorch and TensorFlow